

# XIAOFENG TAN

Shenzhen, Guangdong, China

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## RESEARCH INTERESTS

RLHF, RLVR, AIGC, 3D Human Modeling, Preference Optimization, Diffusion Models, Multimodal Learning.

## EDUCATION

**Southeast University (SEU)**, Nanjing, China

Sept 2024 – Present

CSRankings (2025–2026): AI #14 · CV #41 · ARWU #89

M.Sc. in Computer Science [PALM Lab](#), supervised by Prof. [Hongsong Wang](#)

GPA: **88.6**/100

First-Class Academic Scholarship & Tencent-SEU Scholarship (top 1%); Summer Camp Rank 2

**Shenzhen University (SZU)**, Shenzhen, China

Sept 2020 – Jun 2024

CSRankings (2025–2026): AI #12 · CV #31 · ARWU 151–200

B.E. in Computer Science & B.Sc. in Mathematics (*Dual Degree*)

GPA: **88.9**/100 · **4.07**/4.50

Outstanding Graduate Representative (1/300+) and graduation speaker; featured in [SZU's 2024 Admissions Guide](#)

## SELECTED PUBLICATIONS

\* indicates equal contribution or first author (student.)

### Published

**When Policy Entropy Constraint Fails: Preserving Diversity in Flow-based RLHF via Perceptual Entropy**

Xiaofeng Tan, Jun Liu, Bin-Bin Gao, Yuanting Fan, Xi Jiang, Chengjie Wang, Hongsong Wang, Feng Zheng

[NeurIPS'26](#) · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A\*)

[Spotlight](#) (292/30,709 submissions, 0.95%)

[\[arXiv\]](#) [\[Project\]](#)

**BiMoGen: Bidirectional Motion-Text Generation via Unified Masked Discrete Diffusion**

Wanjiang Weng, Yongliang Wu, Xiaofeng Tan, Xingyu Zhu, Wenbo Zhu, Hongsong Wang

[NeurIPS'26](#) · *Advances in Neural Information Processing Systems* · Poster (CCF-A, CORE-A\*)

**EasyTune: Efficient Step-Aware Fine-Tuning for Diffusion-Based Motion Generation**

Xiaofeng Tan\*, Wanjiang Weng\*, Haodong Lei, Hongsong Wang

[ICLR'26](#) · *International Conference on Learning Representations* (CCF-A, CORE-A\*)

[\[PDF\]](#) [\[Project\]](#) [\[Openreview\]](#)

**ReAlign: Text-to-Motion Generation via Step-Aware Reward-Guided Alignment**

Wanjiang Weng\*, Xiaofeng Tan\*, Junbo Wang, Guo-Sen Xie, Pan Zhou, Hongsong Wang

[AAAI'26](#) · *AAAI Conference on Artificial Intelligence* (CCF-A, CORE-A\*)

[\[PDF\]](#) [\[Project\]](#) [\[Code\]](#)

**FG-Diff: Frequency-Guided Diffusion Model with Perturbation Training for Skeleton-Based VAD**

Xiaofeng Tan, Hongsong Wang, Xin Geng, Liang Wang

[TIP'26](#) · *IEEE Transactions on Image Processing* (CCF-A, SCI-Q1)

[\[Project\]](#) [\[PDF\]](#) [\[Code\]](#)

**SoPo: Text-to-Motion Generation Using Semi-Online Preference Optimization**

Xiaofeng Tan, Hongsong Wang, Xin Geng, Pan Zhou

[NeurIPS'25](#) · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A\*)

[\[Project\]](#) [\[PDF\]](#) [\[Code\]](#)

**Fuzzy Granule Density-Based Outlier Detection with Multi-Scale Granular Balls**

Can Gao (Supervisor), Xiaofeng Tan\*, Jie Zhou, Weiping Ding, Witold Pedrycz

[TKDE'25](#) · *IEEE Transactions on Knowledge and Data Engineering* (CCF-A, SCI-Q1)

[\[PDF\]](#) [\[Code\]](#)

### Preprints

**MotionRFT: Unified Reinforcement Fine-Tuning for Text-to-Motion Generation**

Xiaofeng Tan, Wanjiang Weng, Hongsong Wang, Fang Zhao, Xin Geng, Liang Wang

*Under Review (Submitted to TPAMI) (Under Second-Round Review)*

[\[arXiv\]](#) [\[Project\]](#) [\[Code\]](#)

Dual-Grained Policy Optimization for Hallucination Reduction in Flow-based RL

Xiaofeng Tan, Jun Liu, Yuanting Fan, Bin-Bin Gao, Xi Jiang, Xiaochen Chen, Jinlong Peng, Chengjie Wang, Hongsong Wang, Feng Zheng

Under Review (Submitted to AAAI 2027)

[Project] [arXiv] [Code]

RESEARCH EXPERIENCE

|                                                                                                                                                                                 |                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Tencent Inc., Shenzhen, China<br>Research Intern @ Hunyuan<br>Research Direction: RLHF, RLVR, AIGC                                                                              | Mar 2026 – Present<br>Mentor: Jiaxin Lin                                                    |
| University of California, Merced, Merced, CA (Remote)<br>Remote Visiting Student @ Vision and Learning Lab (VLL)<br>Research Direction: Computer Vision                         | Apr 2026 – Present<br>Mentor: Ming-Hsuan Yang                                               |
| Tencent Inc., Shenzhen, China<br>Research Intern @ YouTu Lab<br>Research Direction: RLHF, RLVR, AIGC                                                                            | Jul 2025 – Mar 2026<br>Mentor: Jun Liu<br>Works: Consistent-RFT, PEC-RFT                    |
| Singapore Management University, Singapore (Remote)<br>Remote Visiting Student @ Language and Vision Lab (LV Lab)<br>Research Direction: RLHF, Motion Generation                | Jul 2024 – Mar 2025<br>Mentor: Pan Zhou<br>Works: SoPo (NeurIPS'25), ReAlign (AAAI'26)      |
| Southeast University, Nanjing, China<br>Master Student @ PAttern Learning and Mining (PALM) Lab<br>Research Direction: RLHF, Motion Generation, Action Recognition<br>MotionRFT | Apr 2024 – Present<br>Mentor: Hongsong Wang<br>Works: FG-Diff (TIP'26), EasyTune (ICLR'26), |
| Shenzhen University, Shenzhen, China<br>Undergraduate Research Assistant @ Computer Vision Institute<br>Research Direction: Outlier Detection, Anomaly Detection                | Oct 2022 – Oct 2023<br>Mentor: Can Gao<br>Works: MGBOD (TKDE), TWCOD (IJAR)                 |

SERVICE

Conference Reviewer: ICML 2026 (Gold Reviewer), NeurIPS 2026, AAAI 2027, ICLR 2027  
Journal Reviewer: Transactions on Machine Learning Research (TMLR), International Journal of Data Science and Analytics (IJDSA)

HONORS & AWARDS

|                                     |                                              |      |
|-------------------------------------|----------------------------------------------|------|
| Tencent-SEU Scholarship             | Southeast University                         | 2025 |
| First-Class Academic Scholarship    | Southeast University                         | 2025 |
| Outstanding Graduate Representative | Shenzhen University (Graduation Speech)      | 2024 |
| Featured in Admissions Guide        | Shenzhen University (as Outstanding Student) | 2024 |